

Week 9 Cheat Sheet

Priority-Aware Multi-Drone Coordination

1. Problem Setup

Scenario:

- Multiple drones delivering medical supplies (organs, blood, etc.)
- Single shared landing pad → bottleneck
- Only one drone can land at a time

Key Idea:

Multi-Agent Coordination + Shared Resource + Conflicting Objectives

This is an SMG (Social Mini-Game):

- Shared constraint (landing pad)
- Multiple agents
- Requires coordination to avoid conflict

2. Priority Model

Each drone is assigned a **priority score**:

$$P_i = w_1 C_i + w_2 T_i + w_3 D_i + w_4 A_i$$

Where:

- C_i = cargo criticality (organ > blood > meds)
- T_i = time sensitivity
- D_i = distance to hospital
- A_i = patient acuity

Important:

- P_i is continuous (not discrete)
- Can be **ambiguous** (similar priorities)

3. Bottleneck / SMG Structure

Landing pad = shared resource

Only one agent can occupy resource at a time

Examples:

- Doorway
- Intersection
- Narrow hallway

Core challenge:

Who goes first?

4. Greedy vs Global Strategy

Greedy Policy

$$\pi_{\text{greedy}} = \arg \max_i P_i$$

- Always choose highest priority next
- Myopic (no future planning)

Global / Lookahead Policy

$$\pi^* = \arg \min \sum_i w_i \cdot \text{delay}_i$$

- Plans over time horizon
- Considers future arrivals

Key Insight:

Local optimal $\neq$ Global optimal

5. Example Conflict Scenario

- Drone A: organ (high priority), far away
- Drone B: blood (medium), very close
- Drone C: organ (critical), just launched

Greedy: wait for A

Better: let B land first $\rightarrow$ clears pad

6. Coordination Mechanisms

Rule-Based

- Fixed priority ordering
- Simple but rigid

Negotiation (LLM / Protocol)

Each drone shares:

- ETA
- cargo type
- urgency

Decision:

yield / proceed

Important:

- Happens **before** conflict
- Avoids real-time latency issues

7. VIP / Priority Override

- Organ drone = VIP
- Others yield

Key Question:

Should VIP ever wait for system efficiency?

Tradeoff:

Individual priority vs System throughput

8. Evaluation Metrics

(1) Total Weighted Delay

$$J = \sum_i w_i \cdot (t_i^{\text{land}} - t_i^{\text{arrival}})$$

(2) Priority Inversion Rate

Inversion = #(lower priority goes first)

(3) Pad Utilization

$$\text{Utilization} = \frac{\text{active landing time}}{\text{total time}}$$

(4) Patient Outcome Proxy

- Score based on delivery time vs urgency

9. Key Tradeoffs

- Fairness vs Efficiency
- Priority vs Throughput
- Local vs Global optimality
- Autonomy vs Coordination

10. Class Connections

- SMG (doorway / intersection)
- Multi-agent coordination
- Scheduling problem
- Game theory / incentives
- Centralized vs decentralized control

11. One-Line Takeaways

- Bottlenecks create coordination problems
- Priority alone is insufficient
- Lookahead improves global performance
- Communication enables better outcomes